Q3

3. $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

$A^* = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$

$A^*A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$

$\det(xI - A^*A) = (x-1)^2(x-2) - (x-1) - (x-1)$

$= (x-1)((x-1)(x-2) - 2)$

$= (x-1)(x)(x-3)$

So the eigenvalues of A^*A are 1 and 3 (0 is because of kernel)

$\Sigma = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3} & 0 \end{pmatrix}$

Finding eigenvectors for:

λ_1:

let $(A^*A - 1I)v_1 = 0$

$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix} v_1 = 0$

$v_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

λ_3:

let $(A^*A - 3I)v_2 = 0$

$\begin{pmatrix} -2 & 0 & 1 \\ 0 & -2 & 1 \\ 1 & 1 & -1 \end{pmatrix} v_2 = 0$

$v_2 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$

let $v_3 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$, which ensures linear independence of $\{v_1, v_2, v_3\}$

$V = \left(\frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \frac{v_3}{\|v_3\|} \right)^T = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & \end{pmatrix}^T$

Then, find U:

$u_1 = \frac{1}{(1)} A v_1 = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$

$u_2 = \frac{1}{(\sqrt{3})} A v_2 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$

$U = \begin{pmatrix} u_1 & u_2 \end{pmatrix}$

$A = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3} & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & -1/\sqrt{3} \\ 2/\sqrt{6} & 1/\sqrt{6} & 1/\sqrt{6} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$

$1/\sqrt{3} \quad 1/\sqrt{3} \quad -1/\sqrt{3}$

$2/\sqrt{2} \quad 1/\sqrt{2} \quad 1/\sqrt{2}$